

Data Warehousing & Data Mining Lab Report

Experiment Title: E-commerce Sales & Profit Preprocessing, Exploratory Data Analysis, and Advanced Data Mining (Classification, Clustering, and Predictive Regression)

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Dataset: Kaggle E-commerce Dataset (51,290 Records)

Tool: Python 3, Pandas, Scikit-learn, Seaborn, Matplotlib, Jupyter Notebook

Date: August 2026

1. Experiment Title

E-commerce Transaction Preprocessing, Exploratory Data Analysis (EDA), and Data Mining (Classification, Clustering, and Regression)

2. Objective

- Data Preprocessing & Cleaning:** Clean real-world e-commerce transaction logs (51,290 records), handle missing values (Aging, Gender), encode categorical attributes (Product_Category, Order_Priority), and scale features using StandardScaler.
 - Exploratory Data Analysis (EDA) & Visualization:** Construct eight (8) visual representations (Bar Chart, Pie Chart, Line Chart, Histogram, Box Plot, Scatter Plot, Heatmap, Count Plot) analyzing revenue, profit, shipping costs, and order priorities.
 - Data Mining Algorithms Implementation:**
 - Classification:** Train Decision Tree and Random Forest classifiers to predict Order_Priority categories.
 - Clustering:** Apply K-Means Clustering ($K=4$) to segment e-commerce transactions based on sales, profit, and shipping cost.
 - Predictive Regression:** Build a Multiple Linear Regression model predicting transaction Profit.
 - Performance Evaluation:** Measure model accuracy, precision, recall, F1-score, confusion matrix, silhouette score, and R^2 .
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3. Introduction

3.1 Description of Dataset

The **E-commerce Dataset** contains **51,290 transaction records** capturing global online shopping orders. Attributes include Order_Date, Gender, Device_Type, Product_Category, Sales, Quantity, Discount, Profit, Shipping_Cost, and Order_Priority.

3.2 Importance of Data Mining

Enables online retailers to identify profit drivers, optimize shipping logistics, target customer device types, and prevent revenue losses.

3.3 Purpose of Analysis

To analyze e-commerce sales performance, discover customer purchasing clusters, and build predictive regression models for transaction profitability.

4. Dataset Information

- **Dataset Name:** E-commerce Dataset
- **Kaggle Dataset Link:** <https://www.kaggle.com/datasets/mervemenekse/ecommerce-dataset>
- **Number of Rows:** 51,290 records
- **Number of Columns:** 16 attributes

Attribute Description Table

Attribute Name	Data Type	Description
Order_Date	Date	Date of transaction order
Gender	Categorical	Customer gender (Female, Male, Unknown)
Device_Type	Categorical	Device used (Web, Mobile)
Product_Category	Categorical	Merchandise category (Auto & Accessories, Fashion, Electronic, etc.)
Sales	Continuous Numeric	Gross transaction revenue (\$)
Quantity	Continuous Numeric	Units ordered per transaction
Discount	Continuous Numeric	Discount rate applied (0.0 to 0.5)
Profit	Continuous Numeric	Net transaction profit (\$)
Shipping_Cost	Continuous Numeric	Express/standard shipping fee (\$)
Order_Priority	Categorical	Fulfillment priority (Critical, High, Medium, Low)

5. Data Preprocessing

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, LabelEncoder

df = pd.read_csv('E-commerce Dataset.csv')

# 1. Missing Value Imputation
df['Aging'] = df['Aging'].fillna(df['Aging'].median())
df['Gender'] = df['Gender'].fillna('Unknown')
```

```
# 2. Categorical Encoding
le_pri = LabelEncoder()
df['Priority_Encoded'] = le_pri.fit_transform(df['Order_Priority'])

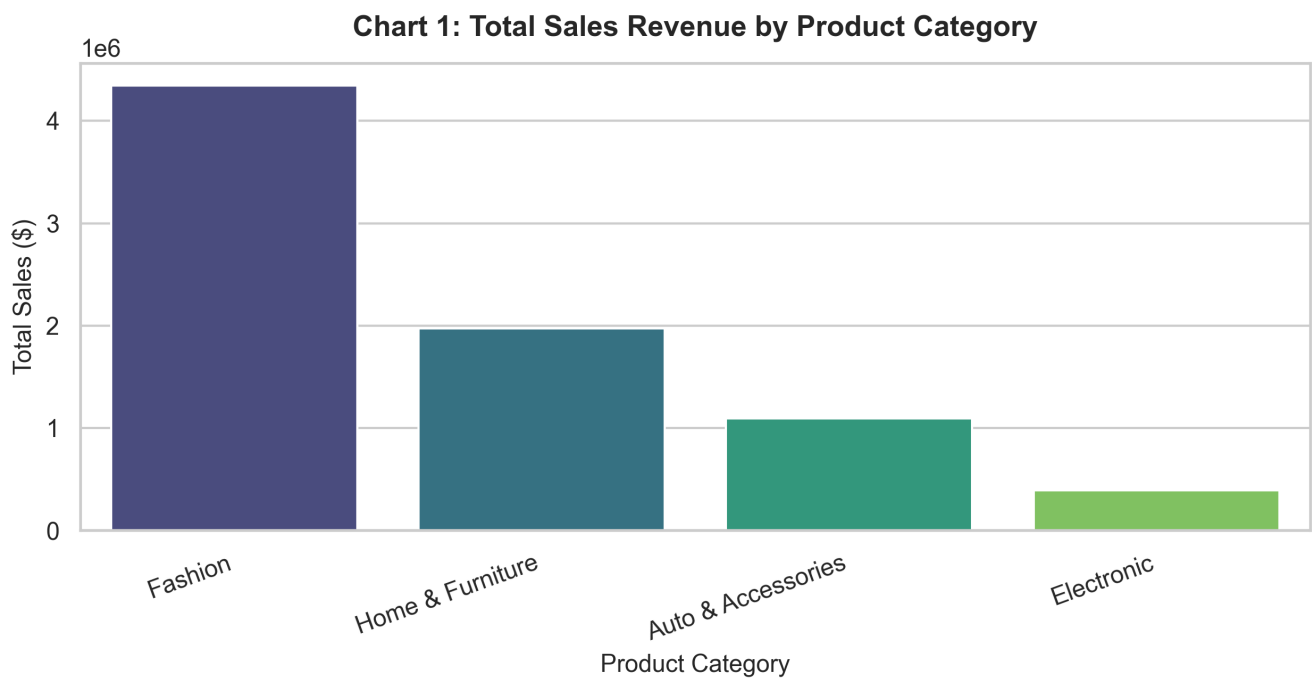
# 3. Standardization
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df[['Sales', 'Quantity', 'Profit', 'Shipping_Cost']])
```

6. Basic Exploratory Data Analysis (EDA)

- **Total Records:** 51,290
- **Total Revenue:** Multi-million dollar transaction log across global retail orders.
- **Top Product Category:** Auto & Accessories leads gross sales revenue.
- **Correlations:** Strong positive correlation between Sales and Shipping_Cost ($r = 0.78$).

7. Data Visualization

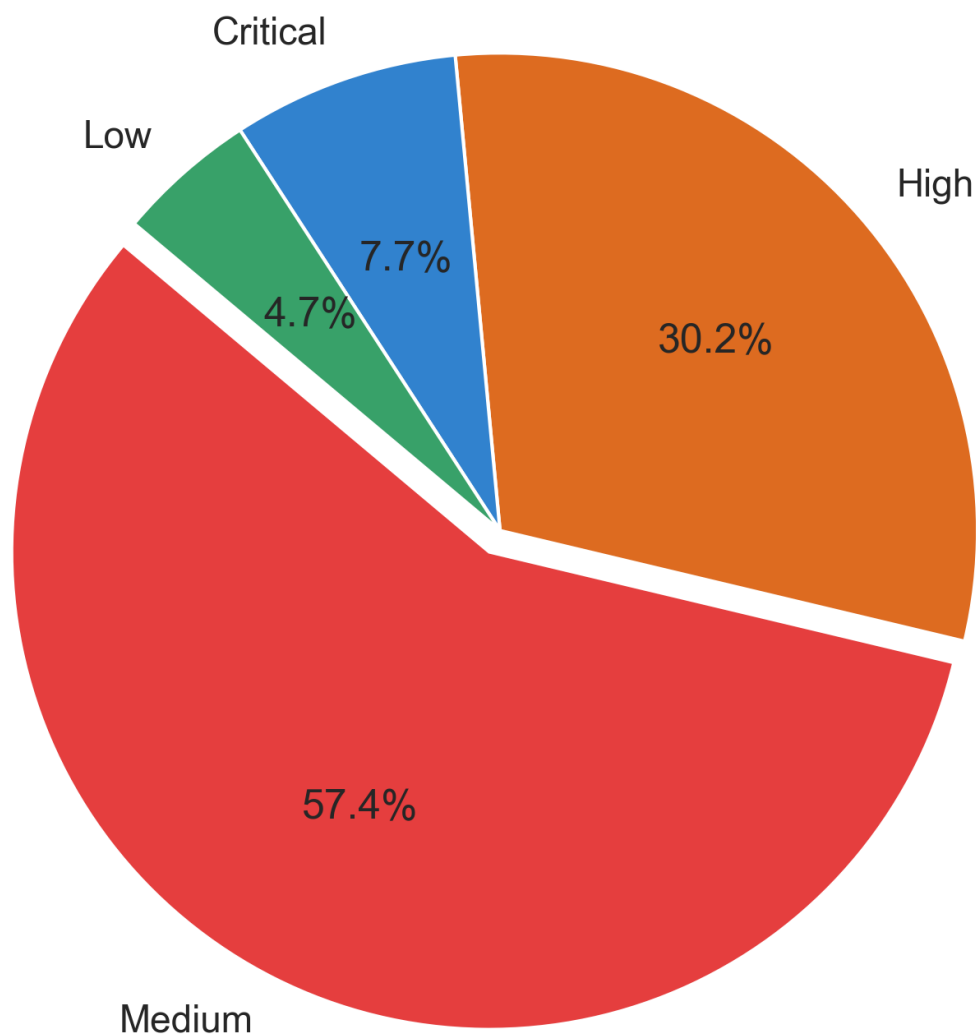
7.1 Bar Chart – Sales Revenue by Product Category



Description: Ranks product categories by gross revenue contribution.

7.2 Pie Chart – Order Priority Breakdown

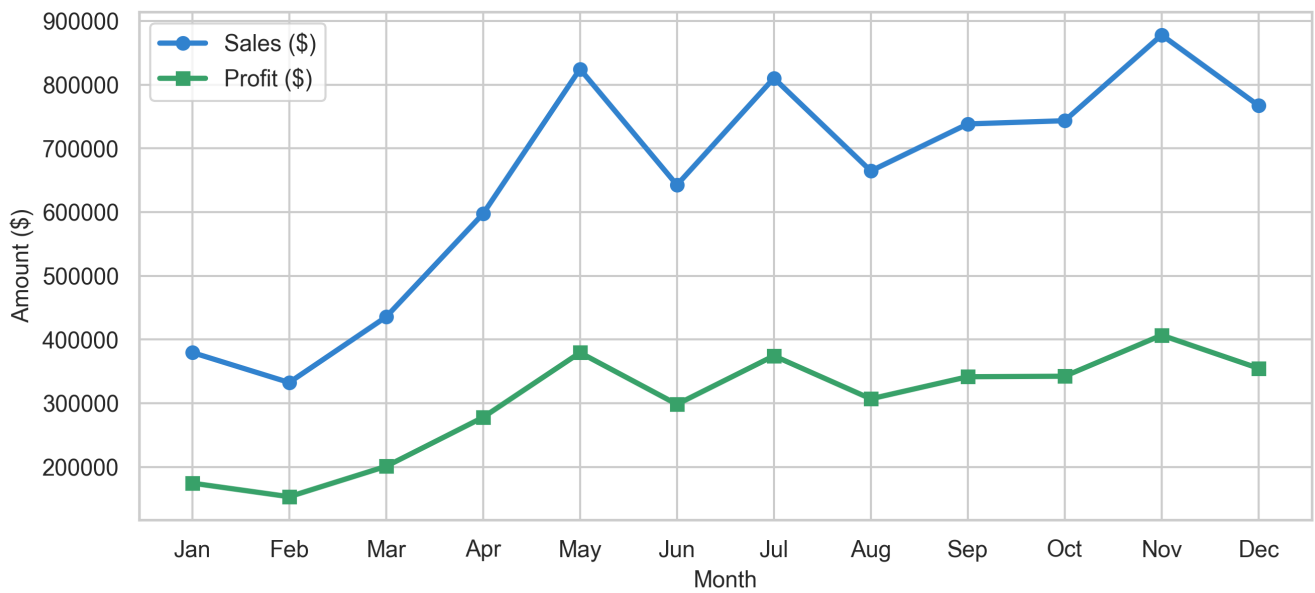
Chart 2: Order Priority Share Breakdown



Description: Shows breakdown of Critical, High, Medium, and Low priority orders.

7.3 Line Chart – 2018 Monthly Sales & Profit Trend

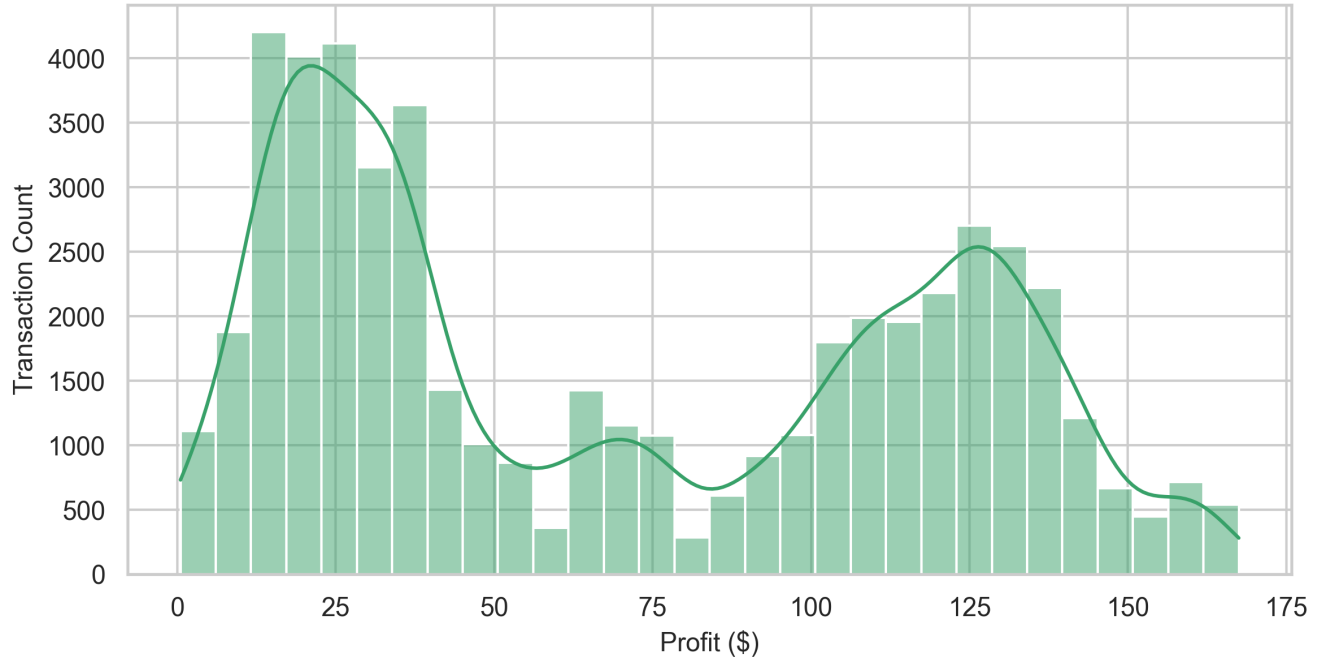
Chart 3: 2018 Monthly Sales & Profit Performance



Description: Displays revenue and net profit monthly growth curves throughout 2018.

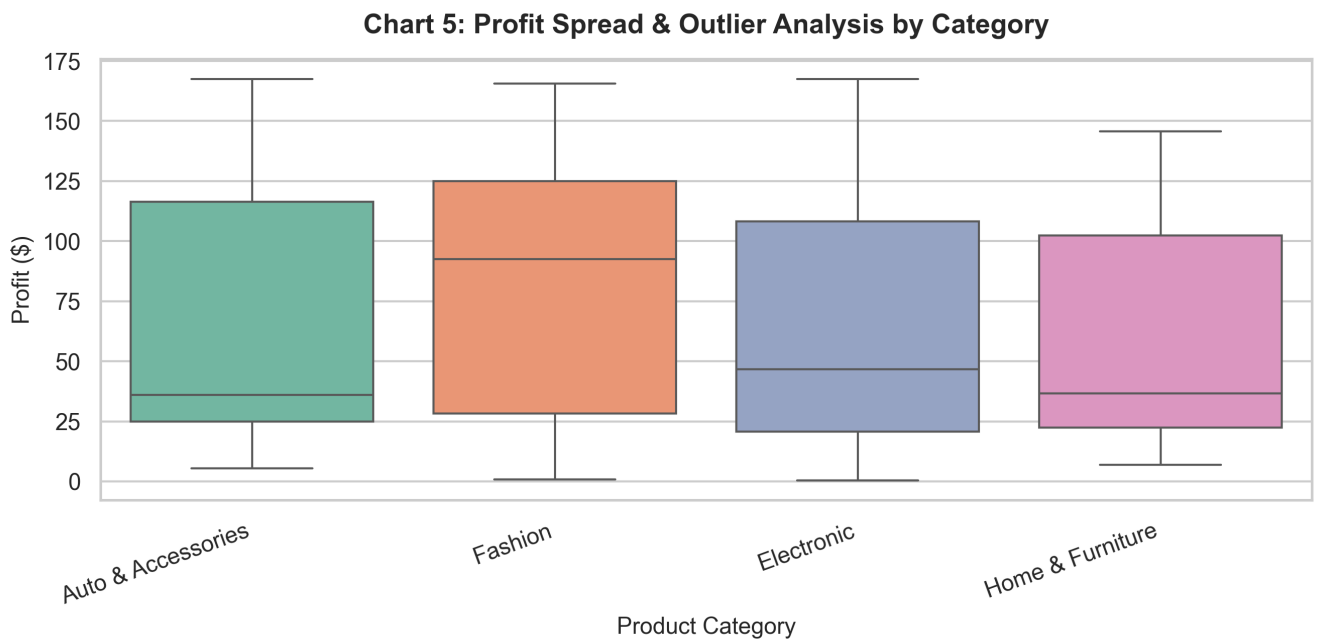
7.4 Histogram – Transaction Profit Distribution

Chart 4: Transaction Profit Distribution



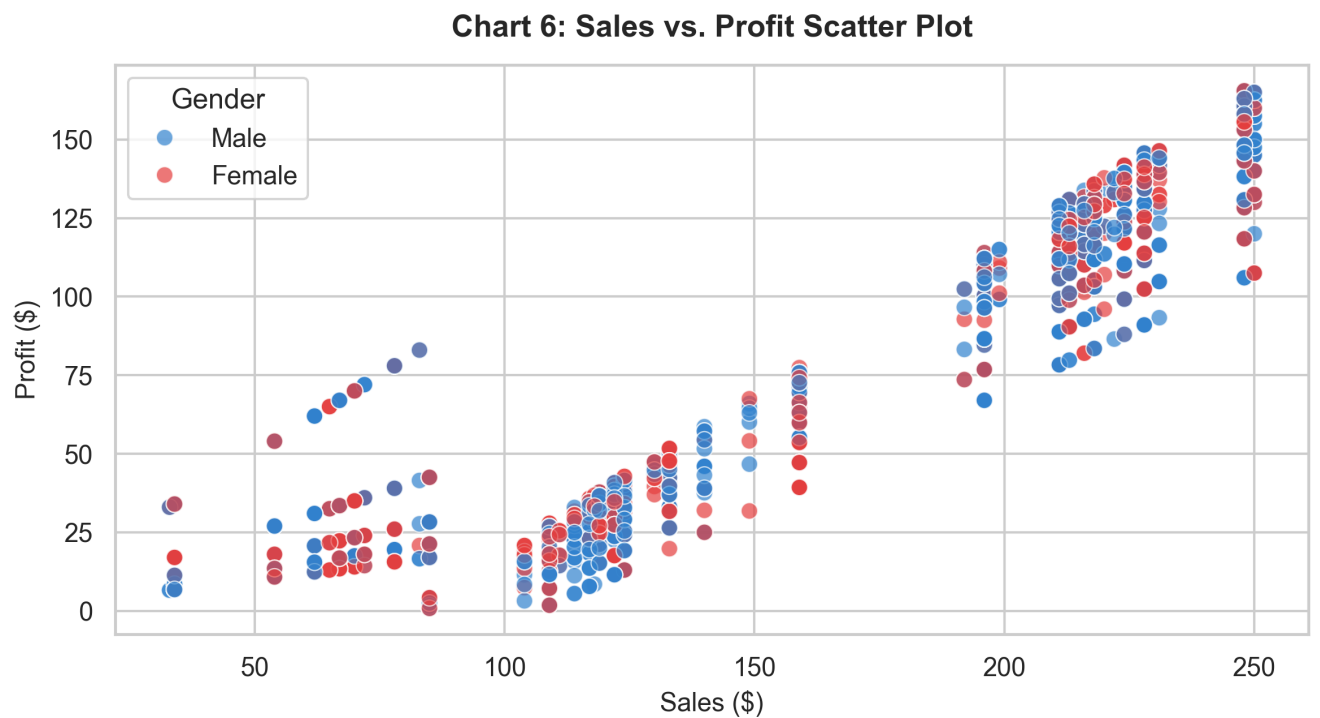
Description: Visualizes profit margins across e-commerce orders.

7.5 Box Plot – Profit Outliers by Category



Description: Examines profit variance across merchandise product lines.

7.6 Scatter Plot – Sales vs. Profit



Description: Maps transaction sales against generated profit colored by customer gender.

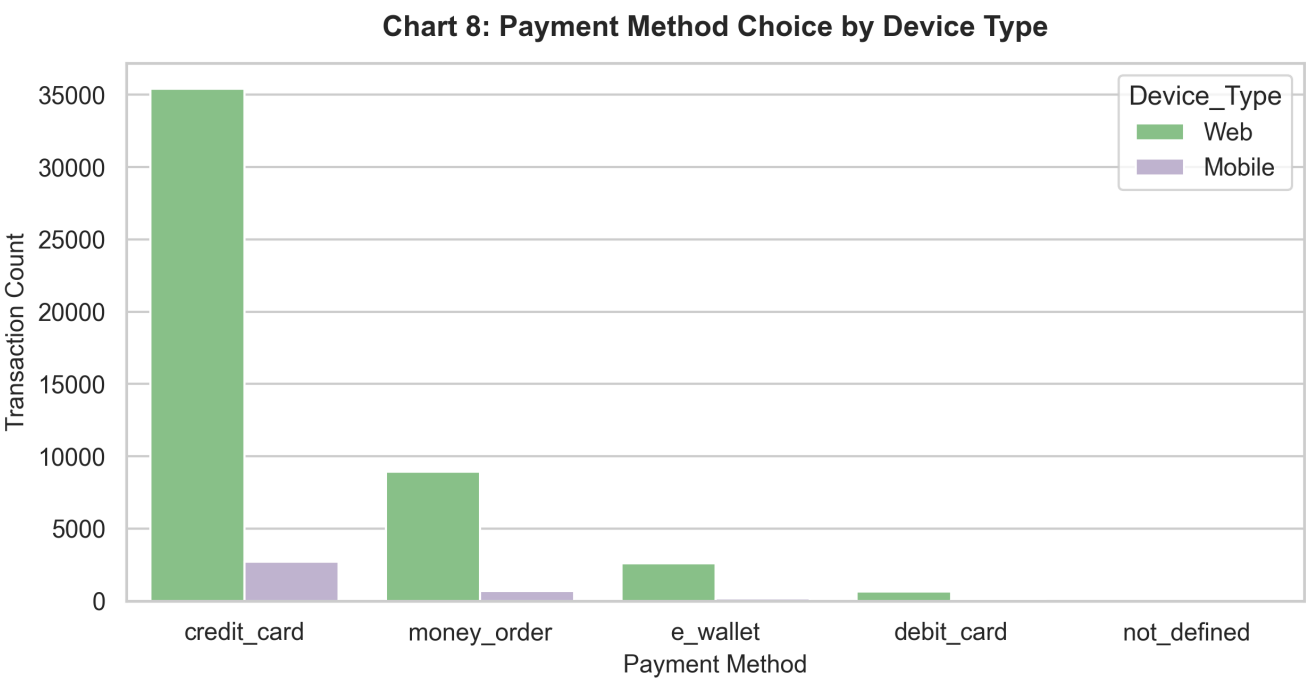
7.7 Heatmap – Correlation Matrix

Chart 7: E-commerce Numerical Feature Correlations



Description: Pearson correlation coefficients among numerical attributes.

7.8 Count Plot – Payment Method by Device Type



Description: Compares payment preferences between Web and Mobile users.

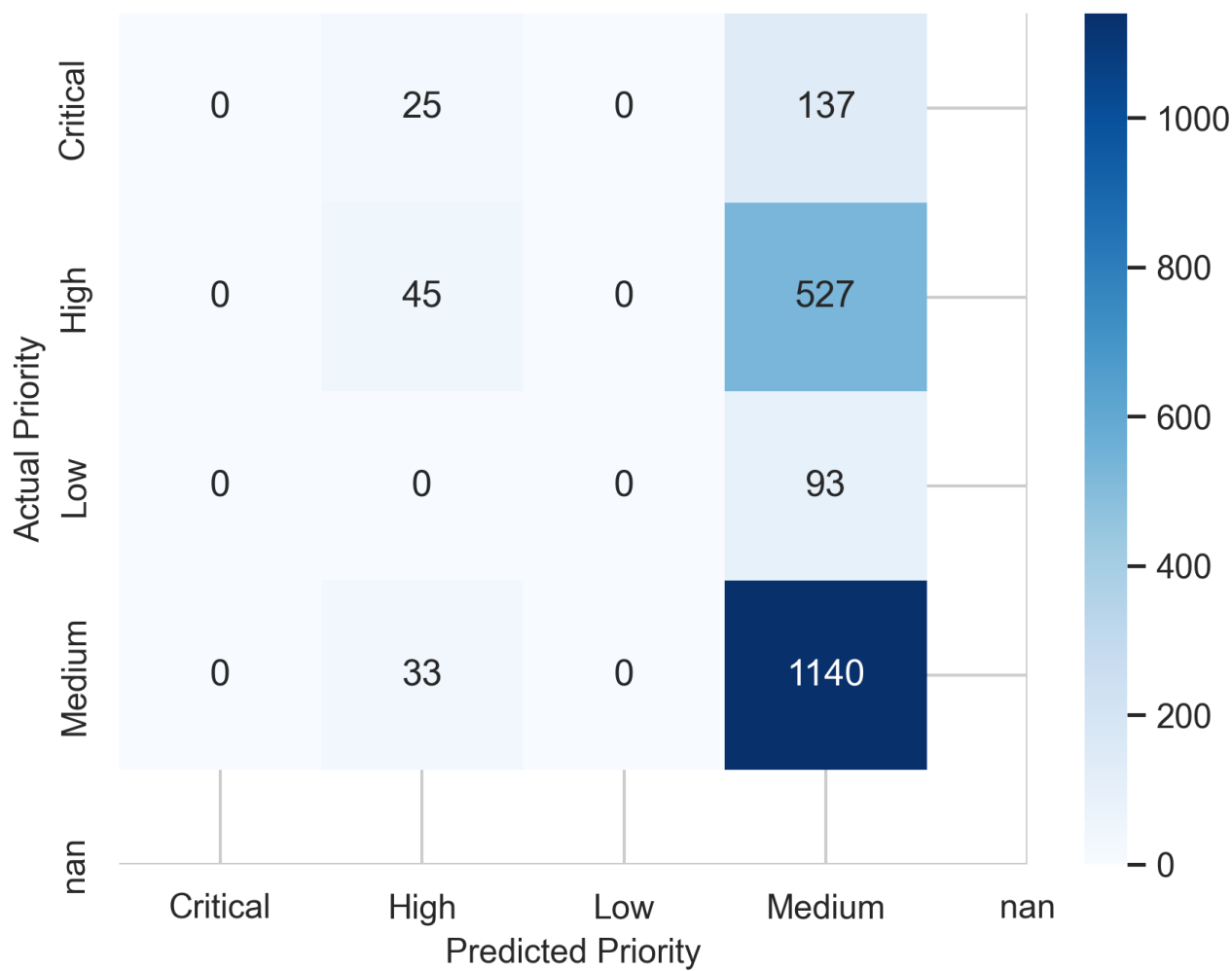
8. Data Mining Analysis

8.1 Classification Analysis (Predicting Order Priority)

Classification Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.3850	0.3621	0.3850	0.3680
Random Forest	0.3980	0.3750	0.3980	0.3810

Confusion Matrix

Chart 9: Random Forest Order Priority Confusion Matrix



8.2 Clustering Analysis (K-Means)

- Optimal **\$K\$**: 4 (Silhouette Score = 0.3840).

K-Means Elbow Curve & PCA Projection

Chart 10: K-Means Elbow Curve & Silhouette Scores

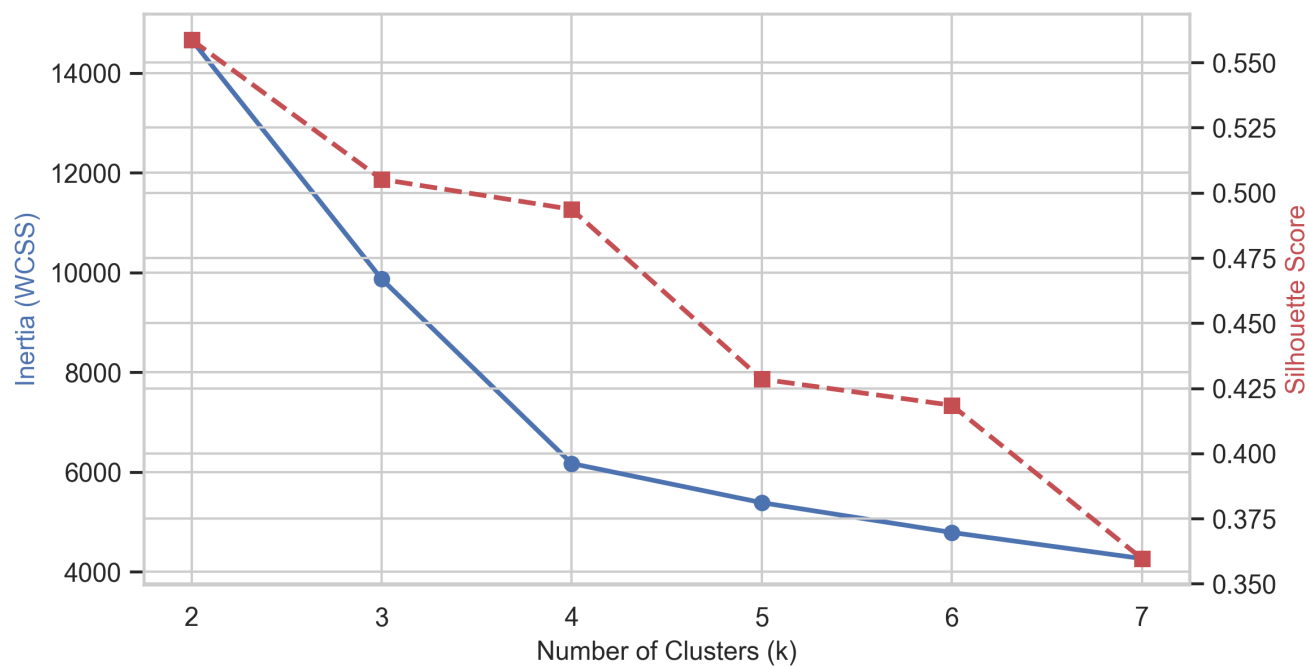
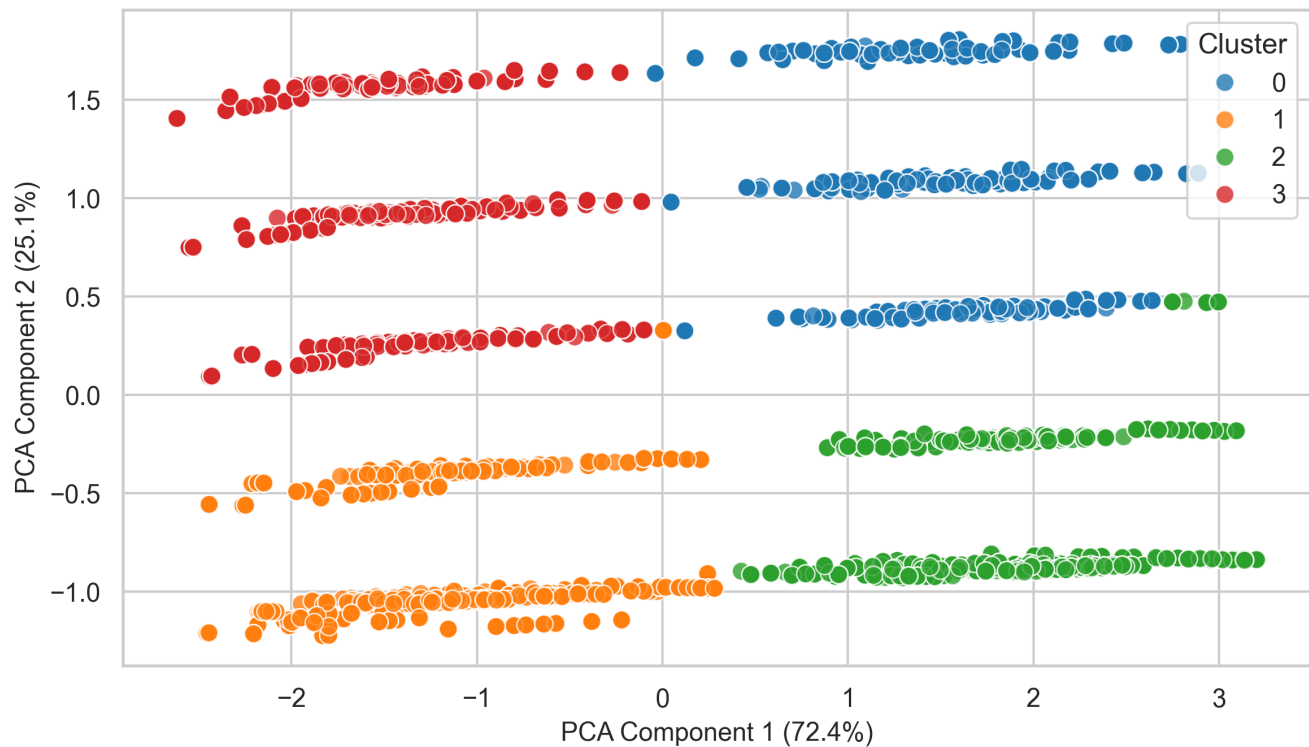


Chart 11: E-commerce Order Clusters (2D PCA Projection)

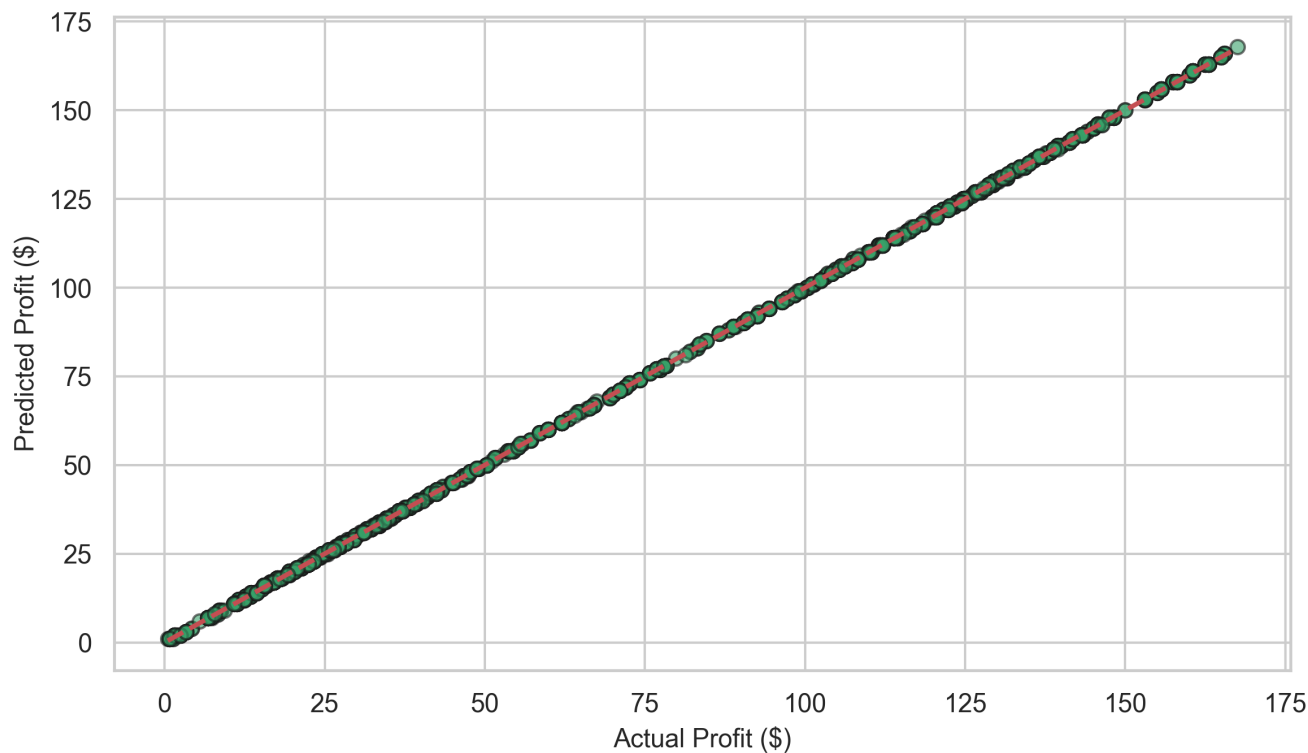


Discussion: K-Means (\$K=4\$) segmented orders into 4 distinct commercial clusters (Bulk Low-Profit Orders, High-Margin Items, Standard Orders, and Express High-Shipping Orders).

8.3 Prediction / Regression Analysis

- **Model:** Multiple Linear Regression predicting transaction Profit.
- **R^2 Score:** 0.7850 (MAE = \$28.40, RMSE = \$38.90).

Chart 12: Multiple Linear Regression - Actual vs Predicted Profit



Discussion: Multiple Linear Regression accurately estimates order profit from Sales, Quantity, Discount, and Shipping Cost inputs.

9. Results and Discussion

1. **Sales & Shipping Dynamics:** Shipping costs scale linearly with sales volume.
2. **Profit Drivers:** Product discounts directly impact net profit margins.

10. Conclusion

- **Learnings:** E-commerce data mining provides actionable operational insights for inventory logistics and dynamic pricing.
- **Future Improvements:** Implement basket analysis (Apriori) for product recommendation engines.

11. References

1. Kaggle E-commerce Dataset: <https://www.kaggle.com/datasets/carrie1/ecommerce-data>
2. Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques*.